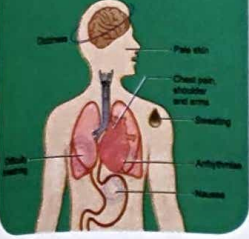


CHAPTER

23



Hridroga

Learning Objectives

- ⇒ To learn the concept of Hridroga
- ⇒ To learn the Nidana Panchaka of Hridroga in Ayurveda literatures
- ⇒ To frame out the probable Samprapti Ghataka based on the progression of the disease
- ⇒ To enlist the types of Hridroga
- ⇒ To know the Lakshana of each type of Hridroga
- ⇒ To know about Vishishta Lakshana of Hridroga
- ⇒ To understand the different Upadrava (complications) of Hridroga
- ⇒ To learn the prognosis of the Hridroga

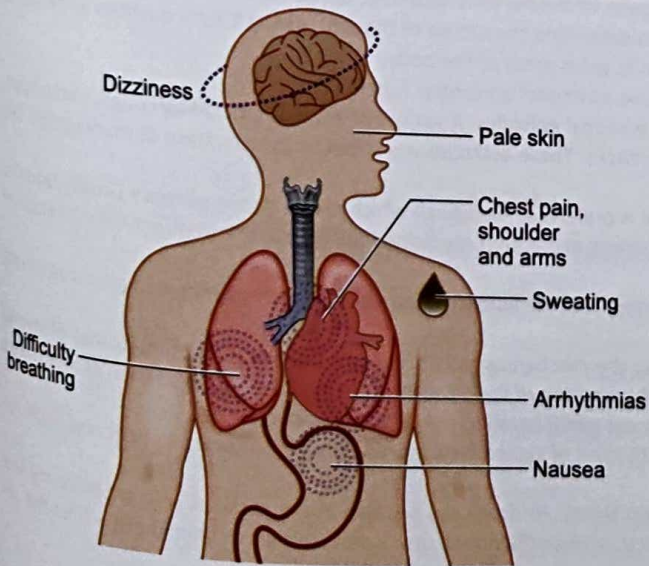
Introduction

हृदो रोगो हृद्रोगः।
कुर्वन्ति हृदये वाधां हृद्रोगं तं प्रचक्षते॥
हृदये बाधां कुर्वन्ति, तं हृद्रोगं प्रचक्षते

Madhu, Ma.Ni. – 29/2

Su.Ut. – 43/4

Dalhana, Su.Ut. – 43/4



and which is leading to discomfort in *Hridaya* (chest) region is termed as *Hridroga*. This is a disease which involves *Rasa Dhatu* in the body, hence *Rasavaha Srotas* is afflicted. As *Hridaya* is the *Srotomula* of *Pranavaha Srotas* too, we can predict the involvement of *Pranavaha Srotas* in due course of time. *Sanga* and *Vimargagamana* is the type of *Srotodushti* involved in the whole manifestation of the disease. There is a possibility of affliction of *Pranavata*, *Udanavata*, *Vyanavata*, *Sadhaka Pitta*, *Avalambaka Kapha*. There may be affliction of *Ojas*, *Manas* too.

As far as the description of *Hridaya* is concerned, it is having three words – *Hri*, *Da* and *Ya*. It indicates continuous and repeated receiving and sending. *Hridroga* is the term which indicates the disturbance in receiving and sending of the nutrients.

Nidana of Hridroga

अत्युष्णगुर्वन्नकषायतिक्तभ्रमाभिघाताभ्यशनप्रसङ्गैः॥१॥

संचिन्तनवेंगविधारणैध हृदरामयः पञ्चविधः प्रदिष्टः॥ Ma.Ni. – 29/1

व्यायामतीव्रणातिविरेकबस्तिचिन्ताभयन्त्रासमदाभिचाराः॥

सूर्यामसम्धारणकर्षणानि ढोगकर्तुनि तथाऽभिघातः। Ca.Ci. – 26/77

वेगाघातोष्णरक्षाप्रेरतिमानोपसेवितैः।
विरुद्धाध्यशनाजीर्णैरसात्यैश्चापि भोजनैः॥३॥

Su.Ut. – 43/3

Figure 67: Pratyatma Lakshana of Hridroga

As per the various definitions of *Hridroga* (see Figure 67), we can learn that any deviation in the function of *Hridaya*

- ❖ Excess consumption of *Ushna Virya*¹ substances
- ❖ Excess consumption of substances which are heavy to digest²
- ❖ Excess consumption of substances having astringent and bitter taste³
- ❖ Excess activities which lead to fatigue⁴ or excess exercise
- ❖ Trauma⁵ to the body
- ❖ Excess consumption of substances⁶ before the digestion of previously ingested meal
- ❖ Excess sexual activity⁷
- ❖ Excess thinking⁸
- ❖ Suppression of the natural urges⁹
- ❖ Excess consumption of medicines which are of strong potency¹⁰
- ❖ Excess purgation and enema¹¹
- ❖ Mental stress, fear¹²
- ❖ Other diseases¹³

G
K

- 1 When we speak the effect of Rasa, it acts upto the level of tongue and oral cavity, whereas the effect of Virya encompasses not only the alimentary canal, even the intracellular compartment of human body can be considered. In other words when an individual consumes Ushna Virya substances such as chilli or mustard oil, we can have effect on any part of the body, but when it involves the region of the heart, he may be presented with cardiac manifestations.
- 2 When an individual consumes substances which are heavy to digest such as black gram items, that will lead to the over nourishment in the body. It is being mentioned that when an individual consumes excess carbohydrates and they are not required for the metabolic activities, that extra glucose will be converted and stored as glycogen and fat. It is being said that who consume excess overnourishing substances, he will be having the higher risk factor of development of cardiac disease.
- 3 Excess consumption of astringent and bitter taste is said to be responsible for the vitiation of Vata Dosha in the body.
- 4 The person who indulges in excess physically exhausting activities lead to fatigue. In most of the occasions, these kinds of activities will cause the depletion of carbohydrates and this in turn will mobilise the stored fat into circulating fat. This may become the risk factor for the development of ischemic heart disease.
One more hypothesis we can put forth is when the person is doing excess fatigue producing activities, it puts more pressure on the dominant chamber of heart i.e. left ventricle producing left ventricular hypertrophy.
- 5 In this situation, the term trauma refers to physical trauma. This kind of the trauma will lead to the excess blood loss and thus will be responsible for the manifestation of hypovolemic shock. This creates more burden on cardiac activity.
- 6 This etiological factor is leading to the overdistension of the stomach. This may be responsible for the commencement of vomiting. This is initiated by activation of vomiting centre and this is accompanied by both vagal and sympathetic afferent nerve fibres mediated by 5th, 7th, 9th, 10th and 12th cranial nerves. The people who are at risk of the development of heart disease, tachyarrhythmia is one of the phenomenon for the manifestation of the heart disease. This mechanism will not be leading to above said pathophysiology, when a person has the habit of the frequent habit of over consumption of the food substances.
- 7 Excess sexual activity is responsible for the activation of autonomic nervous system at different phases of sexual act. When there is activation of sympathetic nervous system, there will be increased heart rate, increased cardiac output, increased pulse rate etc. Frequent sexual intercourse may lead to increased cardiac output leading to extra burden on left ventricle, dominant chamber of the heart. Myocardial muscles receive blood supply only during diastolic phase of the heart.
- 8 Each thought process involves simultaneous signals in many portions of cerebral cortex, thalamus, limbic system and reticular formation of brain stem. A thought results from a pattern of stimulation of many parts of the nervous system at the same time. The stimulated areas of limbic system, thalamus and upper reticular formation are believed to determine the nature of thought, giving it such qualities as pleasure, displeasure, pain, comfort, crude modalities of sensation, localisation to gross areas of the body.
Thus, the nature of the negative thought process continuously can have an impact on cardiac function.
- 9 This cause is explained by the accurate understanding of situational syncopal episodes. A variety of activities including cough, micturition, defecation are associated with susceptible individuals for syncopal attacks. These activities are initiated by activation of manoeuvres like Valsalva, straining and coughing.
- 10 In earlier times, consumption of medicines through alimentary canal is preferred. Medicines which are of strong potency usually leads to the inflammation of stomach in particular. Frequent consumption of strong and potent medicines will lead to the frequent inflammation of alimentary canal.
Thus, frequent consumption of strong potent medicines will lead to transfer of the fluid from intravascular compartment to gastro intestinal lumen. Thus, the person may have a sequel of hypovolemia.
- 11 In these purificatory procedures, pathological mechanism is similar to the mechanism occurs in the administration of medicines of strong potency. In case of excess purgation or Virechana, there will be the inflammation of the small intestine and the administration of medicated enema (Basti) will cause the inflammation of large intestine. Thus, excess administration may lead to hypovolemia.
- 12 Excess exposure to mental stress and fear are responsible for the activation of sympathetic nervous system, leading to fight and flight response. This may initiate the rhythmic disturbances of the heart.
- 13 Charaka stated the description of Srotomula as Hridaya for above said Srotas. Hridroga can be seen in the patients who are suffering the illnesses afflicting either Pranavaha, Rasavaha Srotas. During the course of the affliction of any Srotas, sooner or later there will be the affliction of Srotomula or origin of Srotas.
As per the western perspective,
 - The person who is suffering from chronic respiratory disease, he may develop the condition called cor pulmonale.
 - In the person who is suffering from rheumatic fever, he will likely to develop the mitral valve prolapse.
 - In the patient of wet beriberi, the person may develop cardiac failure.
 - In the person who is suffering from chronic essential hypertension, he will likely to develop left ventricular hypertrophy.

- Suppression of vomiting¹⁴
- Etiological factors causing presence of Ama^{15,16}

Samprapti of Hridroga

दूषयित्वा रसं दोषा विगुणा हृदयं गताः।
हृदि बाधां प्रकुर्वन्ति हृद्रोगं तं प्रचक्षते॥2॥

Ma.Ni. – 29/2; Su.Ut. – 43/-

- First there will be vitiation of *Dosha*
- There will be vitiation of *Rasa*
- Abnormal *Rasa* enters the *Hridaya* (chest) region
- Discomfort in the chest region will be manifested.

Samprapti Ghataka of Hridroga

- **Dosha:** *Vyanavata, Sadhaka Pitta, Avalambaka Kapha*
- **Dushya:** *Rasa*
- **Srotas:** *Rasavaha* and *Pranavaha Srotas*
- **Srotodushti Prakara:** *Sanga* and *Vimargagamana*
- **Agni:** *Jatharagnimandya* and *Rasadhatvagnimandya*
- **Ama:** *Jatharagnimandyajanya* and *Rasadhatvagnijanya*
- **Udbhava Sthana:** *Amashaya*
- **Sanchara Sthana:** *Sarva Sharira* as *Rasa Dhatu* affliction is observed
- **Vyakta Sthana:** *Hridaya*

14 The process of vomiting as follows:

1. A deep breath
2. Raising of hyoid bone and larynx to pull the upper oesophageal sphincter open
3. Closing of glottis to prevent vomitus flow into the lungs
4. Lifting of the soft palate to close the posterior nares
5. Strong downward contraction of diaphragm along with simultaneous contraction of all abdominal muscles
6. Squeezing of stomach between diaphragm and the abdominal muscles resulting in enhanced intragastric pressure.
7. Relaxation of lower oesophageal sphincter leading to expulsion of gastric contents to the exterior.

As far as heart disease is concerned, when the person suppresses the vomiting, this may lead to

- Autonomic nervous system improper activation, when there is sympathetic nervous system is activated, that will lead to tachycardia which may further lead to poor cardiac activity. When there is parasympathetic nervous system is activated, there is a possibility of bradycardia which may lead to poor cardiac output.
- When there is more severe contraction of diaphragm and if the person is having the risk factors for the ischemia, he may likely to develop inferior wall myocardial infarction.

15 Following hypothesis can be put forth to understand the pathological understanding in this mechanism of Ama formation.

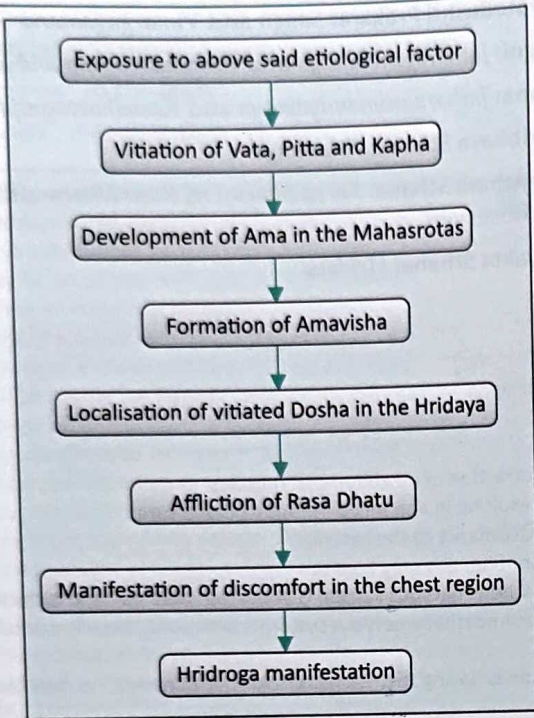
1. Alternate nutrient supply bypassing the alimentary canal. This means due to the absence of adequate nutrient supply through alimentary canal, the accumulated fat in the body in different parts gets mobilised and this will lead to the circulation of triglycerides. If these are get stuck in the coronary vessel, he may likely to get cardiac manifestation.
2. Formation of free radical in the body: This is formed due to causes such as
 - Inflammatory damage
 - Chemical carcinogenesis
 - Atherosclerosis
 - Cellular aging
 - In ionising radiation
 - In chemical toxicity etc.
3. Free radicals are formed within the mitochondrial inner membrane in the presence of cytochrome oxidase. These free radicals are
 - Superoxide oxygen (O₂⁻) – one electron
 - Hydrogen peroxide (H₂O₂) – two electrons
 - Hydroxyl radical (OH⁻) – three electrons

4. These are partially reduced intermediate species between O₂ and H₂O. These will cause degradation of proteins, DNA damage and cytoskeletal damage.

- 16 When the person is continuously exposed to the aetiological factors leading to the presence of Ama in the body, refers to the presence of macromolecules in the alimentary canal will remain in the stomach for more than 2 hours secondary to closure of pyloric antrum.
- As the human being consumes more amount of carbohydrates during meal which are macromolecules in nature, are leading to the infection because of the presence of microbes in the stomach.
 - As there is poor functioning of gastric juice (abnormal secretion of mucus, poor functioning of hydrochloric acid (normal pH 1.8 to 3.5; in case of abnormal diet pattern, pH >3.5); poor conversion of inactive pepsinogen into active pepsin); inflammation of stomach occurs resulting in the formation of inflammatory markers.
 - Inflammatory markers are proteins in nature. These are responsible for increased movement of fluid in the stomach and gastric mucosal damage occurs later. The conversion of macromolecules into intermediate or micromolecules will be decreased.
 - Macromolecules or intermediate molecules will enter the interstitial compartment of stomach, which later enter the lymphatic system and later enters into the superior vena cava via the junction between left internal jugular vein and left subclavian vein.
 - These macromolecules are heavier in weight, they move slower in the circulatory system. When the macromolecules pumped into the aorta, these molecules will move at the late phase of ejection of blood from left ventricle. The coronary vessels are situated at the root of aorta, the macromolecules will enter the coronary arteries, thereby the person who has considerable degree of Ama, likely to afflicted with heart disease.

- ❖ Rogamarga : Abhyantara, Bahya and Madhyama
- ❖ Vyadhi Svabhava: Ashukari
- ❖ Sadhyasadhya: Krichrasadhya to Asadhya

Samprapti chart



Samanya Lakshana of Hridroga

वैवर्ण्यमूर्च्छज्वरकासहृक्काश्वासास्यवैरस्यतृषाप्रमोहाः छर्दिः कफोत्क्ले-

शरुजोऽरुचिश्च हृद्रोगजाः स्युविविधास्तथाऽप्ये॥

- ❖ Discolouration of the body parts¹⁷
- ❖ Altered state of consciousness¹⁸
- ❖ Raised body temperature¹⁹
- ❖ Cough²⁰
- ❖ Hiccup²¹
- ❖ Breathing discomfort²²
- ❖ Abnormal taste perception in the mouth
- ❖ Thirst²³
- ❖ Confusion²⁴
- ❖ Vomiting
- ❖ Nausea²⁵
- ❖ Pain²⁶
- ❖ Tastelessness²⁷

Sankhya Samprapti of Hridroga

5 types – as per Charaka, Madhavakara, Vagbhata

1. Vataja
2. Pittaja
3. Kaphaja
4. Sannipataja
5. Krimija

4 types: as per Sushruta,

1. Vataja
2. Pittaja

वैवर्ण्यभेदेन ग्लानिहर्षरीक्ष्यस्नेहा

- 17 As per this reference, we should not restrict the word Vaivarnya only to discolouration. It includes the texture of the skin too. Excess oedema or reddish discolouration or hardness of the swelling can be considered as Vaivarnya. In this context of Hridroga, we have to consider cyanosis or pallor or oedema on the different parts, both feet in particular.
- 18 This is attributable to the initiation of syncopal attack. This occurs due to the poor pumping activity of the heart.
- 19 This is attributable to the infection and inflammatory mechanism of the heart.
- 20 This is observed in the person who is presented with left sided heart failure either atrium or ventricle. This is usually associated with expectoration of pink and frothy sputum.
- 21 This clinical symptom is seen when the person is afflicted with involvement of inferior wall of the heart which is in direct contact with diaphragm. Hiccup is usually seen when there is excess irritation of the diaphragm.
- 22 This is the cardinal symptom of the heart disease. The grading of breathless in cardiac disease is breathlessness on exertion, paroxysmal nocturnal dyspnoea, orthopnea and Cheyne Stokes respiration. Refer the Nidana Prakasha written by the author for the above description in detail.
- 23 3 mechanisms are responsible for the manifestation of thirst.
 1. Decrease in extracellular fluid volume and arterial pressure
 2. Increased secretion of angiotensin II
 3. Dryness of the mouth and oesophagus
 Among the above mentioned causes, hypovolemia is considered to be one of the mechanism responsible for thirst.
- 24 This clinical symptom is attributable to the poor cerebral blood supply which happens as a consequence of syncopal attack.
- 25 Vomiting and nausea are the symptoms observed in an individual who has cardiac compromise. This is seen in the individuals who are suffering from right sided heart failure. These symptoms are attributable to backward failure extending from right side of the heart till portal vein.
- 26 This is attributable to poor blood supply to the myocardial musculature. The clinical conditions which are responsible for tachycardia such as anemia or hypertension or excess exercise will lead to poor coronary circulation in the people who are having the risk of development of atherosclerosis.
- 27 Tastelessness can be attributable to increased secretion in the stomach and small intestine which is a sequel of backward failure which in turn leads to poor gastric secretion.

Chapter 23

3. Kaphaja
4. Krimija

Vataja Hridroga:

शोकोपवासव्यायाम शुष्काल्प भोजनः।
वायुराविश्य हृदयं जनयत्युत्तमां रुजम्॥३०॥
वैपथ्यवैष्टनं स्तम्भः प्रमोहः शून्यता देरः।
हृदि वातातुरे रूपं जीर्णं चात्यर्थवेदना॥३१॥ Ca.Su. – 17/30,31
हृद्व्यभवद्रवशोषभेद- स्तम्भाः समोहाः पवनाद्विशेषः।
Ca.Ci. – 26/79

आयस्यते मारुतजे हृद्रोगं तुद्यते तथा।
निर्मथ्यते दीर्यते च स्फोटयते पाठयतेऽपि च॥३२॥
Ma.Ni. – 29/3; Su.Ut. – 43/6

वातेन शूल्यतेऽत्यर्थं तुद्यते स्फुटतीव च॥३३॥
भिर्यते शुष्यति स्तब्ध हृदय शून्यताद्रवः।
अकस्माद्दीनता शोको भयं शब्दासहिष्णुता॥३४॥
बेपथुर्वेन मोहः श्वासरोधोऽल्पनिद्रता। A.Hr.Ni. – 5/39,40

• If the person is having grief, fasting, excess exercise, excess intake of ununctuous and dry substances, inadequate or little amount of food intake

• Vata Dosha is vitiated and it reaches Hridaya region

- Severe pain in the chest
- Involuntary movements
- Cramps
- Stiffness of the chest region
- Altered state of consciousness
- Sensation of emptiness
- Severe pain which worsens after the digestion of food

As per Sushruta,

- Chest discomfort simulates pulling type of pain,
- Crushing type of pain
- Pricking or piercing type of pain
- Splitting type of pain

As per Vagbhata,

- Severe pain in the chest which simulates breaking or tearing type of pain
- Cracking type of pain
- Stiffness
- Sudden stoppage of function of heart
- Sense of emptiness

Pittaja Hridroga

पित्ताम्ललवणक्षारकटुकाजीर्णभोजनैः।
सोधात हृदि पित्तं प्रकुप्यति॥३२॥
हृदि तिक्तता बक्त्रे तिकाम्लोद्गिरणं क्लमः।
पित्ताम्लमूर्च्छा भ्रमः खेदः पित्तहृद्रोगलक्षणम्॥३३॥ Ca.Su. – 17/32,33

पित्ताम्लमूर्च्छा भ्रमो मूर्च्छा दाहः स्वेदोऽम्लकः क्लमः॥३४॥
छर्दनं चाम्लपित्तस्य धूमकः पीतता ज्वरः।
Ca.Ci. – 26/79
Ma.Ni. – 29/4; Su.Ut. – 43/7

पित्ताम्लमूर्च्छा भ्रमो मूर्च्छा दाहः स्वेदोऽम्लकः क्लमः॥३४॥
छर्दनं चाम्लपित्तस्य धूमकः पीतता ज्वरः।
A.Hr.Ni. – 5/41

- Excess intake of Ushna Virya (hot potency) substances
- Excess intake of Amla (sour), Lavana (salt), Katu (pungent) Rasa substances
- Excess intake of Kshara substances such as vinegar
- Excess food intake in a state of Ajirna (indigestion)
- Excess intake of alcohol
- Anger state
- Exposure to hot sunlight
- Pitta Dosha gets vitiated in Hridaya

As per Charaka, the Lakshana of Pittaja Hridroga:

- Burning sensation in the chest region
- Bitter taste in the mouth
- Tiktamlodgara (bitter or sour eructations or both)
- Fatigue without physical exertion
- Thirst
- Altered state of consciousness
- Giddiness
- Excess sweating
- Darkness in front of the eyes
- Confusion
- Fearfulness
- Raised body temperature
- Yellowish discolouration of body parts

As per Sushruta,

- Thirst
- Local or generalised raise of temperature
- Burning sensation all over the body
- Burning sensation on the sides of the chest
- Chest discomfort without any physical exertion
- Feeling of smoke coming through one's mouth
- Altered state of consciousness
- Excess sweating
- Dryness of mouth

As per Vagbhata:

- Thirst
- Giddiness
- Altered state of consciousness

- ⊗ Burning sensation
- ⊗ Sweating
- ⊗ Fatigue without any physical exertion
- ⊗ Vomiting
- ⊗ Burning sensation in the chest region associated with sour or bitter eructation
- ⊗ Sensation of smoke coming through mouth
- ⊗ Yellowish discolouration of the body parts
- ⊗ Raise of temperature

Kaphaja Hridroga

- अत्यादानं गुरुस्निग्धमचिन्तनमचेष्टनम्।
निद्रासुखं चाभ्यधिकं कफहृद्रोगकारणम्॥34॥
हृदयं कफहृद्रोगे सुप्तं स्तिमितभारिकम्।
तन्द्रारुचिपरीतस्य भवत्यश्मावृतं यथा॥35॥ Ca.Su. - 17/34,35
स्तब्धं गुरु स्यात् स्तिमितं च मर्म कफात् प्रसेकज्वरका सतन्द्राः।
Ca.Ci. - 26/80
गौरवं कफसंस्त्रावोऽरुचिः स्तम्भोऽग्निमार्दवम्।
माधुर्यमपि चास्यस्य बलासावतते हृदि॥36॥ Ma.Ni. - 29/5; Su.Ut. - 43/8
श्लेष्मणा हृदयं स्तब्धं भारिक साङ्गमर्भवत्॥42॥
कासान्निहासद निष्ठीवनिद्रालस्यारुचिज्वरा। A.Hr.Ni. - 5/42

Causes of Kaphaja Hridroga

- ⊗ Excess intake of food
- ⊗ Excess intake of *Guru* (heavy to digest) food substances
- ⊗ Excess intake of *Snigdha* (unctuous) food substances
- ⊗ Not having any thinking at all
- ⊗ Physical inactivity
- ⊗ Excess sleep

Lakshana of Kaphaja Hridroga

According to Charaka

- ⊗ Numbness
- ⊗ Feeling of wet cloth covering whole body
- ⊗ Heaviness of the body parts
- ⊗ Drowsiness
- ⊗ Tastelessness
- ⊗ Feeling of stone kept over one's chest
- ⊗ Stiffness
- ⊗ Excess salivation
- ⊗ Raised body temperature
- ⊗ Cough

As per Sushruta

- ⊗ Heaviness of body parts
- ⊗ Excess salivation from one's mouth
- ⊗ Tastelessness
- ⊗ Stiffness
- ⊗ Loss of appetite
- ⊗ Sweetish taste in the mouth

As per Vagbhata

- ⊗ Stiffness
- ⊗ Heaviness of the chest
- ⊗ Feeling of stone kept over one's chest
- ⊗ Cough
- ⊗ Loss of appetite
- ⊗ Sputum production
- ⊗ Excess sleep
- ⊗ Laziness
- ⊗ Tastelessness
- ⊗ Raised body temperature

Tridoshaja Hridroga

- हेतुलक्षणसंसर्गादुच्यते सान्निपातिकः। Ca.Su. - 17/36
विद्यात्रिदोषं त्वपि सर्वलिङ्गम्- Ma.Ni. - 29/5; Ca.Ci. - 26/80
⊗ Causative factors of all three *Dosha* are seen in this type of *Hridroga*.
⊗ *Lakshana* of *Tridoshaja Hridroga* are mixed *Lakshana* of *Vataja*, *Pittaja* and *Kaphaja Hridroga* altogether.

Krimija Hridroga

- त्रिदोषजे तु हृद्रोगे यो दुरात्मा निषेवते॥36॥
तिलक्ष्मीरगुडादीनि ग्रन्थिस्तस्योपजायते।
मर्मैकदेशे संक्लेदं रसश्चात्योपगच्छति॥37॥
संक्लेदात् क्रिमयश्चास्य भवन्त्युपहतात्मनः।
मर्मैकदेशे ते जाताः सर्पन्तो भक्षयन्ति च॥38॥
तुद्यमानं स हृदयं सूचीभिरिव मन्यते।
छिद्यमानं यथा शस्त्रैर्जातकण्डू महारुजम्॥39॥
हृद्रोगं क्रिमिजं त्वेतैर्लिङ्गैर्युका सुदारुणम्।
त्यरेत जेतुं तं विद्वान् विकारं शीघ्रकारिणम्॥40॥

Ca.Su. - 17/36-40

Ca.Ci. - 26/80

तीव्रार्तितोदं क्रिमिजं सकण्डूम्॥

तीव्रार्तितोदं क्रिमिजं सकण्डूम्॥

उत्क्लेदः धीवनं तोदः शूलं हल्लासकस्तमः।

अरुचिः श्यावनेत्रत्वं शोथश्च क्रिमिजे भवेत्॥

Ma.Ni. - 29/5-6; Su.Ut. - 43/9

कृमिभिः श्यावनेत्रता॥43॥

तमःप्रवेशो हल्लासः शोषः कण्डूः कफलुति।

हृदयं प्रततं चात्र क्रकचेनेव दार्यते॥४४॥
चिकित्सामयं घोरं तं शीघ्रं शीघ्रकारिणम्।

- ✱ Excess intake of Tila (sesamum seeds), milk²⁸, jaggery²⁹ etc
- ✱ Krimija Hridroga is caused by the vitiation of all three Dosha altogether
- ✱ Formation of nodules in the Hridaya (heart) region
- ✱ Rasa becomes sticky in this region of Hridaya
- ✱ These organisms will crawl and eat away the structures of Hridaya.
- ✱ The patient feels as if his heart is being pierced by weapons
- ✱ This condition is associated with itching, pain in the chest region
- ✱ Very severe pricking type of pain
- ✱ Nausea or vomiting or both
- ✱ Darkness in front of the eyes
- ✱ Tastelessness
- ✱ Blackish discolouration around the eyes
- ✱ Swelling

A.Hr.Ni. – 5/43-45

Upadrava of Hridroga

कुमः सादो भ्रमः शोषो ज्ञेयास्तेषामुपद्रवाः।

किमिजे किमिजातीनां लैष्मिकाणां च ये मताः॥७॥

Ma.Ni. – 29/7

- ✱ Fatigue³⁰ without any physical activity
- ✱ Sluggish movement of body parts³¹
- ✱ Giddiness³²
- ✱ Dryness of the body parts³³

Sadhyasadyata of Hridroga

(हृद्रोगः कष्टदः कष्टसाध्य उत्तो महर्षिभिः)

Ca.Su. – 17/36

हृद्रोगं क्रिमिजं त्वेतैर्लिङ्गेषुवा सुदारुणम्।

Ca.Su. – 17/40

There is no specific mentioning of Sadhyasadyata of different subtypes of Hridroga in the literatures. In Hridroga, primary and major Rogamarga afflicted will be Madhyama. Hence, any of the type of Hridroga is by nature is difficult to cure.

Further, Sannipataja Hridroga is considered to be difficult to cure. Krimija Hridroga is considered to be Daruna in nature (Rogabala is more than Rogibala).

Key points of the chapter

- ✱ Hridroga is a spectrum of the diseases where we can observe dysfunction in terms of the Hridaya. This disease involves Rasavaha Srotas primarily and there will affliction of Pranavaha Srotas in due course of time.
- ✱ This is Amashayasamuttha Vikara with Sanga and Vimargagamana as Srotodushti can be observed in this particular disease.
- ✱ Though all Rogamarga affliction can be observed, but Madhyama Rogamarga is the major Rogamarga can be involved during the progression of the disease.
- ✱ As far as the types of Hridroga are concerned, Charaka, Vagbhata and Madhavakara mentioned 5 types of Hridroga
 1. Vataja, 2. Pittaja,
 3. Kaphaja, 4. Sannipataja and
 5. Krimija,
- ✱ Whereas Sushruta mentioned 4 types of Hridroga –
 1. Vataja, 2. Pittaja,
 3. Kaphaja 4. Krimija.
- ✱ As per Sushruta, Tridosha vitiation is seen in Krimija Hridroga.

²⁸ Brucella, pseudomonas and mycobacterium tuberculosis are some of microbes observed in milk.

²⁹ It is being observed that there will be increased bacterial proliferation in stored jaggery and 9250 per gram of fungal load is usually observed in fresh jaggery.

³⁰ If the person is consuming excess amount of jaggery, these microbes will enter the human body easily.

³¹ This is attributable to poor peripheral circulation secondary to reduced cardiac output

³² This can be attributable to poor venous drainage which in turn leads to the bilateral pedal oedema, thus the person may have sluggish movement of body parts secondary to excess fluid accumulation of the body parts

³³ This is attributable to poor circulation to the head and neck leading to the manifestation of syncopal attack. This occurs as a consequence of forward failure i.e. adequate blood will not be pumped into the aorta, so that sufficient blood will not move into cerebral vessels.

³³ This can attributable to the poor peripheral circulation owing to reduced cardiac output. Due to reduced peripheral circulation, sweating etc will not occur and the person may have dryness of the skin on exposure to external environment.

Self-Assessment: Questions for Reviews

Short essays

- ⊗ Explain in brief about Nidana of Hridroga
- ⊗ Explain the Lakshana of Hridroga
- ⊗ Explain Krimija Hridroga
- ⊗ Explain Vataja Hridroga
- ⊗ Explain Upadrava of Hridroga

MCQs

1. The affliction of Vata in Hridroga occurs is
 - a. Samana Vata; b. Vyana Vata;
 - c. Apana Vata; d. Udana Vata
 Answer: (b)
2. As per Sushruta, the types of Hridroga is
 - a. 3; b. 4;
 - c. 5; d. 2
 Answer: (b)
3. Consumption of Guda is responsible for the manifestation of
 - a. Kaphaja Hridroga; b. Pittaja Hridroga;
 - c. Krimija Hridroga; d. Sannipataja Hridroga
 Answer: (c)
4. Major Rogamarga involved in the manifestation and progression of Hridroga is
 - a. Bahya; b. Abhyantara;
 - c. Madhyama; d. All of the above
 Answer: (c)
5. Bhrama is considered to beof Hridroga
 - a. Purvarupa; b. Rupa;
 - c. Arishta; d. Upadrava
 Answer: (d)
6. Which part of the heart is afflicted in the patient suffering from cough with pink frothy sputum?
 - a. Right atrium;
 - b. Right ventricle;
 - c. Left atrium;
 - d. Right atrium & right ventricle
 Answer: (c)
7. When the diagnosis of cor pulmonale is made, which of the following part of the heart is afflicted in the diseased individual?
 - a. Right ventricle and right atrium;
 - b. Left atrium and left ventricle;
 - c. Right and left atria;
 - d. Right and left ventricles
 Answer: (a)

8. Which of the following is not included in acute coronary syndrome?
 - a. Acute myocardial infarction;
 - b. Dissecting aortic aneurysm;
 - c. Pulmonary embolism;
 - d. Congestive cardiac failure
 Answer: (d)

9. Which of the following is used in the diagnosis of cardiac diseases?
 - a. Electro encephalogram;
 - b. Electrocardiogram;
 - c. Electromyogram;
 - d. Nerve conduction study
 Answer: (b)

10. Which of the following is not situated in Hridaya?
 - a. Avalambaka Kapha; b. Sadhaka Pitta;
 - c. Alochaka Pitta; d. Vyana Vata
 Answer: (c)

Chapter resources

- ⊗ <https://www.ganeshdiagnostic.com/blog/complete-guide-to-heart-disease>
- ⊗ Charaka Samhita, Sutra Sthana – 17th chapter – Kiyantah Shirasiya Adhyaya
- ⊗ Charaka Samhita, Chikitsa Sthana -26th chapter – Trimarmiya Chikitsa Adhyaya
- ⊗ Sushruta Samhita, Uttara Tantra – 43rd chapter – Hridroga Pratishedha Adhyaya
- ⊗ Ashtanga Hridaya Nidana Sthana – 5th chapter – Rajayakshmaadi Nidana Adhyaya
- ⊗ Madhava Nidana – 29th chapter – Hridroga Nidana
- ⊗ Textbook of pathology by Harsh Mohan, 5th edition, Jaypee brothers, New Delhi, 2005
- ⊗ Golwalla's medicine for students by Aspi F Golwalla and Sharukh A Golwalla, edited by Milind Y Nadkar, 25th edition, Jaypee Brothers Medical Publishers (P) Ltd, New Delhi, 2017
- ⊗ The Merck Manual of Diagnosis and Therapy by Robert S Porter and Justin L Kaplan, 19th edition, Merck Sharp & Dohme Corp, New Jersey, 2011
- ⊗ Chamberlain's symptoms and signs in clinical medicine by Colin Ogilvie and Christopher C Evans, 12th edition, Butterworth-Heinemann Publishers, 1997
- ⊗ Best & Taylor's physiological basis of medical practice by O P Tandon and Y Tripathi, 13th edition, Wolters Kluwer Health (India) Pvt. Ltd, Gurgaon, 2012
- ⊗ Textbook of Medical physiology by Arthur C Guyton and John E Hall, 11th edition, Saunders Elsevier publications, Philadelphia, 2006